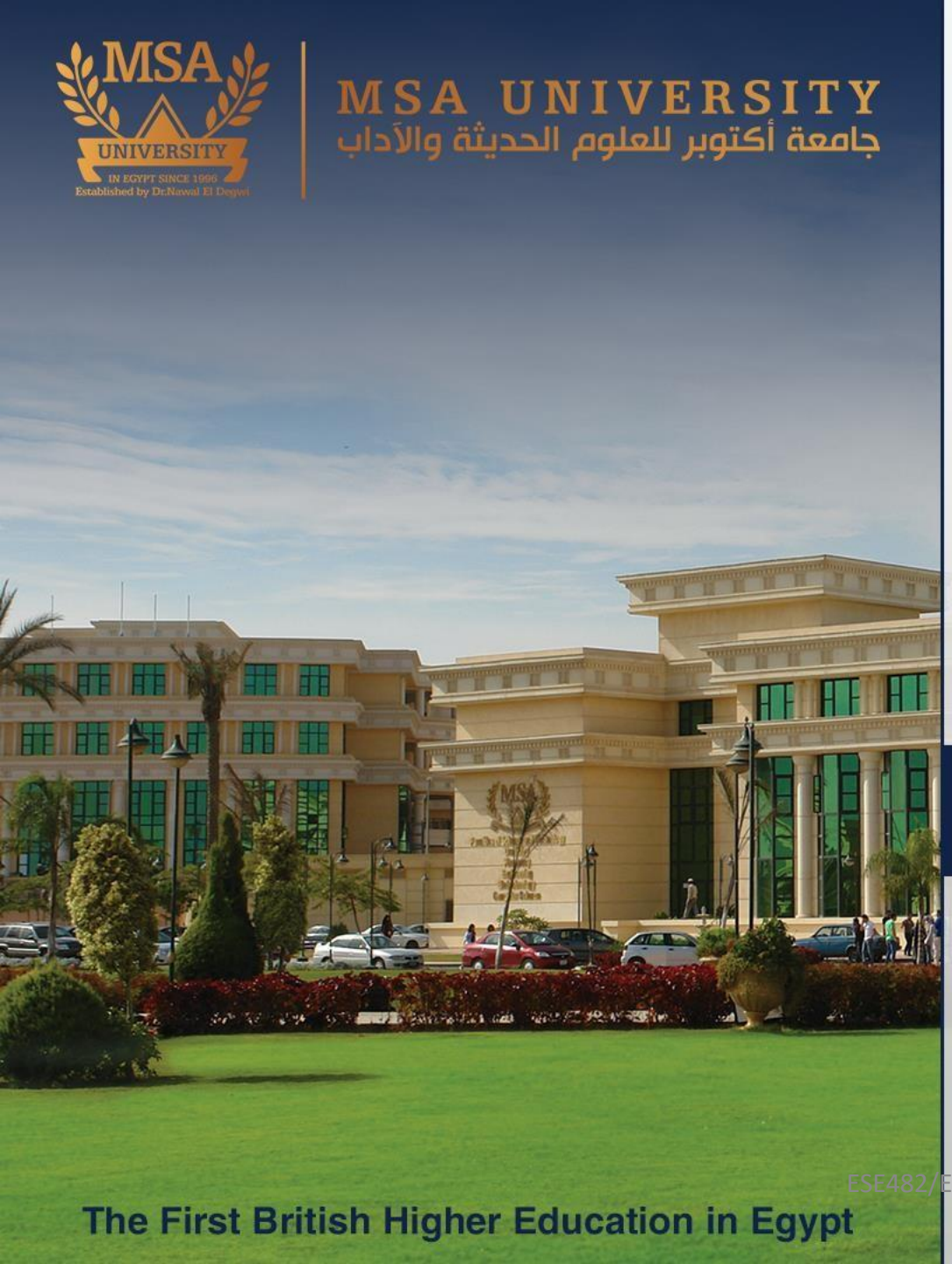


Avionics Navigation Systems

ESE482\ECE4635 - Fall 2026

Lecture 2

Dr. Farah Raad

Lecture Topic

Course Introduction

Chapter 1 : Introduction to Avionics

- Definitions
- Guidance versus Navigation
- Categories of Navigation
- The Vehicle
- Phases of Flight
- Design Trade-offs
- Evolution of Air Navigation
- Integrated Avionics
- Human Navigator



1.1 Definitions

Avionics = Aviation + Electronics

- Used in USA in early 1950's.
- Avionic System / Avionic subsystem: – any system in the aircraft which is dependent on electronics for its operation.

- Fly by wire Flight control system, Fly-by-wire (FBW) is a system that replaces the conventional manual flight controls of an aircraft with an electronic interface.

- Advanced avionics systems can automatically perform many tasks that pilots and navigators previously did by hand.
- For example, an area navigation (RNAV) or flight management system (FMS) unit accepts a list of points that define a flight route, and automatically performs most of the course, distance, time, and fuel calculations.
- Once en route, the FMS or RNAV unit can continually track the position of the aircraft with respect to the flight route, and display the course, time, and distance remaining to each point along the planned route. An auto pilot is capable of automatically steering the aircraft along the route that has been entered in the FMS or RNAV system.

1.1 Definitions

“If an aircraft is flying from Cairo to London, how does it know where it is and where it should go?”

✈ Navigation

Navigation is the process of determining the position and motion (velocity) of a vehicle and determining how to reach a desired destination.

- **Where am I? →**
- **Where am I going? →**
- **How fast am I moving?**

1.1 Definitions

For an aircraft, navigation mainly answers:

Where am I?

and provides information such as:

- Position
- Velocity
- Direction
- Altitude
- Time

➢ **Position**

The position of an aircraft in space can be described using:

- Latitude
- Longitude
- Altitude

➢ **Velocity**

Velocity describes how the aircraft is moving and includes:

- Speed
- Direction of motion

Therefore, a navigation system provides an estimate of the aircraft's current state.

1.1 Definitions

- Navigation data is not used solely to display the aircraft's location to the pilot; it is also transmitted to other systems within the aircraft, such as:

- Flight Control
- Flight Management System
- Engine Control
- Communication Systems
- Crew Displays
- Weapon Control Systems in military aircraft

1.2 Navigation versus Guidance

➢ **Navigation**

Navigation determines: **Where are we ?**

For example, a navigation system may determine that an aircraft is currently at:

Latitude: 30° N , Longitude: 31° E , Altitude: 30,000 ft

➢ **Guidance**

Guidance determines: **Where should we go, and what path should we follow?**

It uses the navigation information together with the desired destination or flight path to determine the path that the aircraft should follow.

✓ For example: **Current Position → Desired Destination**

The guidance system determines the desired direction or trajectory.

➢ **Control**

The control system determines: **How can I make the aircraft follow the desired path?**

The Control is concerned with executing the Guidance's commands by controlling the plane. It generates commands for the aircraft's control systems.

✓ **The basic relationship is: Navigation → Guidance → Control**

- **Navigation tells you where you are.**
- **Guidance tells you where to go.**
- **Control makes the aircraft go there.**

1.3 Categories of Navigation

1. Dead Reckoning

If We know: Starting position, Velocity, Heading, Time. Then, we can estimate the new position. (Errors accumulate over time)

1. Self-contained Navigation

self-contained navigation system determines navigation information using equipment carried onboard the aircraft. It does not require continuous external navigation signals. Examples include:

- **INS — Inertial Navigation System**
- **IRS — Inertial Reference System**

❖ **Advantages**

- Independent of ground stations and satellites
- Can operate over oceans and remote areas
- Does not require external radio signals

❖ **Disadvantage**

Inertial systems are affected by measurement errors. These errors can accumulate with time, resulting in **navigation drift**.

2. Radio Navigation

Radio navigation systems use radio signals transmitted by external stations. Examples include:

- **VOR — Very High Frequency Omnidirectional Range**
- **DME — Distance Measuring Equipment**
- **ADF — Automatic Direction Finder**
- **NDB — Non-Directional Beacon**

✓ The plane receives a signal from a ground station and uses it to know: (Direction, Distance, Bearing, Position) Depending on the system

✓ Radio navigation was extremely important in the development of modern air navigation.

1.3 Categories of Navigation

3. Satellite Navigation

Satellite navigation uses signals transmitted by satellites to determine (Position, Velocity, Accurate time)

The most familiar example is:

- **GPS — Global Positioning System**
- **GNSS — Global Navigation Satellite System**

GPS/GNSS has become a major source of navigation information in modern aircraft.

4. Inertial Navigation

relies on:

- **Accelerometers**
- **Gyroscopes**

It calculates the aircraft's movement without the need for direct external signals.

5. Multisensor Navigation

A very important concept in modern avionics, Instead of relying on a single sensor, we integrate multiple sources of information.

✓ For example: GPS + INS

This yields better performance than relying on either one alone.

1.4 The Vehicle

Navigation systems must be designed according to the characteristics of the vehicle being navigated.

An aircraft moves in three-dimensional space.

- North/South
- East/West
- Up/Down

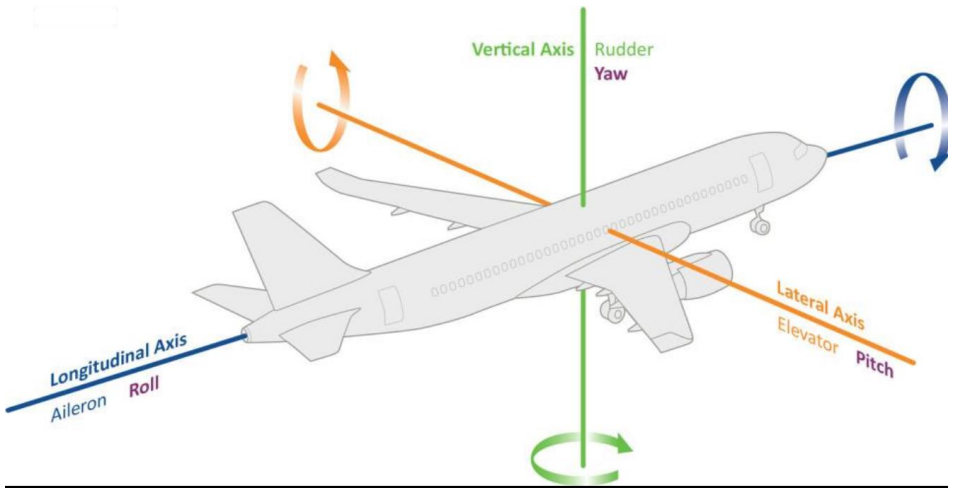
In addition to position and velocity, aircraft navigation systems may need information about the aircraft's **attitude**.

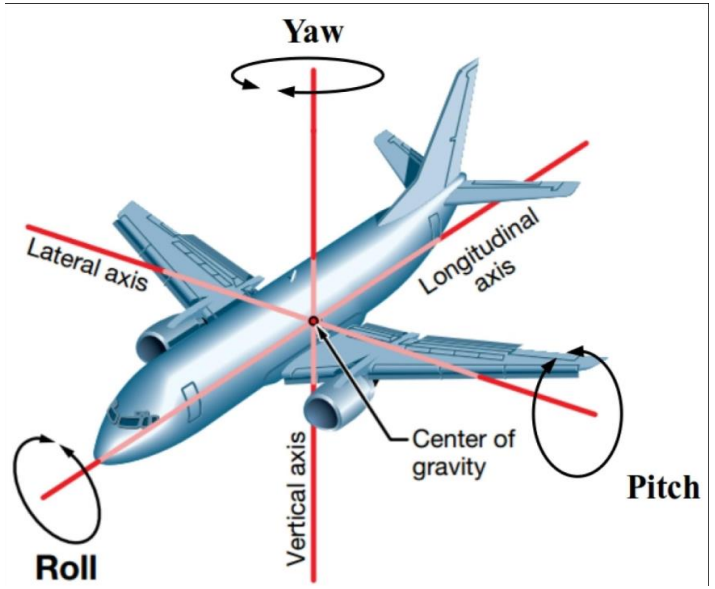
➢ **Aircraft Attitude**

The attitude means the plane's orientation, The three principal attitude angles are:

- **Roll:** Rotation about the longitudinal axis.
- **Pitch:** Rotation about the lateral axis.
- **Yaw:** Rotation about the vertical axis.

These parameters are important because the aircraft's orientation affects its motion and navigation solution.





1.4 The Vehicle

- **Civil Aircraft**

Civil aircraft typically prioritize:

- Safety
- Reliability
- Passenger comfort
- Fuel efficiency
- Accurate navigation
- Air traffic management

- **Military Aircraft**

In addition, they require:

- High accuracy
- High dynamics
- Survivability
- Navigation in GPS-denied environments
- Integration with weapon systems

1.5 Phases of Flight

The navigation requirements of an aircraft change throughout the flight.
The plane doesn't use navigation the same way at every stage of the flight. And each stage has different requirements.
Typical phases include: **Takeoff → Climb → Cruise → Descent → Approach → Landing**

- **Takeoff :** Important information includes:

- Runway guidance
- Heading
- Attitude
- Position
- Altitude

- **Climb :** The aircraft transitions from the airport environment to its planned flight path.

Navigation and guidance systems help maintain the required trajectory.

- **Cruise :** Navigation accuracy becomes particularly important because the aircraft must follow a planned route.

The route may contain several **Waypoints** For example: **Waypoint 1 → Waypoint 2 → Waypoint 3 → Destination**

- **Descent :** The aircraft begins moving toward the destination and must follow the required descent path.
- **Approach and Landing :** High navigation accuracy is required because the aircraft is close to the ground. Depending on the operation, systems such as:

- ILS (Instrument Landing System)
- GNSS
- VOR
- DME

may contribute to navigation and approach guidance.

1.5 Phases of Flight

❑ **Missed Approach**

If landing are not safe, the following is executed:

- Missed approach / Go-around
- The aircraft returns to a designated flight path.

❑ **Surface**

Movement on the airport grounds:

- Taxi
- Runway movement
- Navigation here helps determine the location of aircraft on the airport surface.

❑ **Weather**

Weather can significantly affect navigation requirements. Such as:

Rain, Clouds, Fog, Wind, Storms

1.6 Design Trade-Offs

When designing an avionics navigation system, engineers must balance several competing requirements.
A system may need to provide:

- High accuracy
- High reliability
- Low cost
- Low weight
- Low power consumption
- Small size
- Fast response
- High availability

However, improving one parameter may negatively affect another.

For example:

Higher accuracy may require more sophisticated sensors and processing, increasing cost, weight, and power consumption.

Therefore, navigation system design involves making appropriate **engineering trade-offs**.
The designer must determine which characteristics are most important for the particular aircraft and application.

1.7 Evolution of Air Navigation

Aircraft navigation has developed significantly over time.
A simplified evolution can be represented as:

- **Human Navigator**
- **Visual Navigation**
- **Radio Navigation**
- **Inertial Navigation**
- **Satellite Navigation**
- **Integrated Navigation**
- **Early Navigation**

Early aircraft relied heavily on: Visual references, Maps, Magnetic compasses, Human calculations
The navigator had an important role in determining the aircraft's position and route.

- **Radio Navigation:** The development of radio navigation allowed aircraft to determine direction and distance using signals from ground stations.
- **Inertial Navigation :** Inertial navigation introduced the ability to determine aircraft motion using onboard sensors.

It became particularly important for long-range and military aircraft.

- **Satellite Navigation :** The development of GPS and other GNSS systems provided highly accurate position information using satellite signals.
- **Integrated Navigation :** Modern aircraft combine information from multiple navigation sources.

This provides a more robust and reliable navigation solution.

1.8 Integrated Avionics

Modern aircraft rarely depend on only one navigation source rather than individual (**GPS system, INS system, VOR system)**
Instead, several systems can work together.
For example: **GPS + INS/IRS + Air Data + Radio Navigation**
The information from different sensors and systems can be combined to obtain a better estimate of the aircraft's state.
This concept is often referred to as **Sensor Fusion**

- **Sensor Fusion :** means combining measurements from multiple sensors or navigation systems to produce a more accurate and reliable estimate. (**Integrated Navigation System**)

For example:

- GPS provides position information.
- INS provides continuous motion information.
- Air data systems provide airspeed and altitude information.
- Radio navigation systems provide additional position or range information.

The integrated system can use these sources together rather than relying on only one sensor.

- **Why is integration useful?**

Because every navigation system has limitations.

For example:

- GPS can be affected by signal availability or interference.
- INS can accumulate errors over time.
- Radio navigation depends on available ground infrastructure.
- Combining different systems can reduce the effect of individual system limitations.

1.9 The Human Navigator

In early aviation, the **human navigator** played a major role in aircraft navigation.
The navigator was responsible for tasks such as:

- Determining aircraft position
- Planning the route
- Reading maps
- Calculating headings
- Monitoring navigation information
- Making corrections during flight

✓ With the development of advanced avionics, many of these tasks became automated.

However, the human operator remains important.

✓ The role has changed from performing calculations manually to:

- Monitoring systems
- Managing navigation equipment
- Interpreting information
- Making decisions
- Handling abnormal situations

Therefore:

Automation changes the role of the human operator rather than simply eliminating the human.

Summary

- The key idea of Chapter 1 is that aircraft navigation is not simply about knowing the aircraft's location.

A modern navigation system must:

- Determine the aircraft's state
- Determine the desired route
- Support guidance and control
- Use different navigation technologies
- Integrate information from multiple sensors
- Provide reliable and accurate information throughout all phases of flight
- The development of avionics has moved from human-based navigation toward highly automated and integrated navigation systems, while the human operator continues to play an important role in monitoring and decision-making.

MSA UNIVERSITY
IN EGYPT SINCE 1996
Established by Dr.Nawal El Degwi
MSA UNIVERSITY
جامعة أكتوبر للعلوم الحديثة والآداب

Thank You!

See You next Lectures!!
Any Question?

THE FIRST BRITISH HIGHER EDUCATION IN EGYPT
26th July Mehwar Road Intersection with Wahat Road, 6th of October City, Egypt
Tel: +202383711146 Fax: +20238371543 Postal code: 12451
Email: info@msa.eun.eg Hotline: 16672 Website: www.msa.edu.eg